

Sphere – Levelled UP:

sphere.h	sphere.cpp
1 #ifndef SPHERE_H	1 #include "sphere.h"
2 #define SPHERE_H	2 namespace cs225 {
3 namespace cs225 {	3 Sphere::Sphere()
4 class Sphere {	4 {
5 public:	5 }
6 Sphere();	6
7 Sphere(double r);	7 Sphere::Sphere(double r) {
8 double getRadius();	8 r_ = r;
9 double getVolume();	9 // lots of other logic...
10	10 }
11	11
12	12 double Sphere::getRadius() {
13	13 return r_;
14 private:	14 }
15 double r_;	15
16	16 double Sphere::getVolume() {
17 };	17 return (4 * r_ * r_ * r_ * 3.14159) / 3.0;
18 }	18 }
19	19 }
20 #endif	20 }

Idea: _____ allow us to invoke other constructors to increase code reuse and eliminate copy/pasted code.

sphere.h	sphere.cpp
1 #ifndef SPHERE_H	1 #include "sphere.h"
2 #define SPHERE_H	2 namespace cs225 {
3 namespace cs225 {	3 Sphere::Sphere(double r)
4 class Sphere {	4 {
5 public:	5 // lots of other logic...
6 Sphere(double r=1);	6 }
7 double getRadius();	7
8 double getVolume();	8 double Sphere::getRadius() {
9 private:	9 return r_;
10 double r_;	10 }
11	11
12 };	12 double Sphere::getVolume() {
13 }	13 return (4 * r_ * r_ * r_ * 3.14159) / 3.0;
14	14 }
15 #endif	15 }
	16 }

Stack Memory



example1.cpp			
1	int main() {		
2	int a;		
3	int b = -3;		
4	int c = 12345;		
5			
6	int *p = &b;		
7			
8	return 0;		
9	}		
Location	Value	Type	Name
0xffff00f0	→		
0xffff00e8	→		
0xffff00e0	→		
0xffff00d8	→		
0xffff00d0	→		

example2.cpp	
3	int main() {
4	cs225::Sphere s;
5	cs225::Sphere *p = &s;
6	
7	return 0;
8	}

Location	Value	Type	Name
0xffff00f0 →			
0xffff00e8 →			
0xffff00e0 →			
0xffff00d8 →			
0xffff00d0 →			

Stack Frames

All variables (including parameters to the function) that are part of a function are part of that function's **stack frame**. A stack frame:

- 1.
- 2.

stackframe.cpp			
1	int hello() {	6	int main() {
2	int a = 100;	7	int a;
3	return a;	8	int b = -3;
4	}	9	int c = hello();
5		10	int d = 42;
		11	
		12	return 0;
		13	}

Location	Value	Type	Name
0xffff00f0 →			
0xffff00e8 →			
0xffff00e0 →			
0xffff00d8 →			
0xffff00d0 →			

INSIGHT PUZZLE – What happens here?

sphere-badCreate.cpp			
4	Sphere *CreateUnitSphere() {		
5	Sphere s(1);		
6	return &s;		
7	}		
8			
9	int main() {		
10	Sphere *s = CreateUnitSphere();		
11	double r = s->getRadius();		
12	double v = s->getVolume();		
13	return 0;		
14	}		

Location	Value	Type	Name
0xffff00f0 →			
0xffff00e8 →			
0xffff00e0 →			
0xffff00d8 →			
0xffff00d0 →			

Stack Memory – Solidifying Understanding

stack-array.cpp			
5	int first = 42;		
6	int arr[6];		
7			
8	cout << &(first) << endl;		
9	cout << &(arr[0]) << endl;		
10	cout << &(arr[1]) << endl;		
11	cout << &(arr[2]) << endl;		

Location	Value	Type	Name
0xffff00f0 →			
0xffff00e8 →			
0xffff00e0 →			
0xffff00d8 →			
0xffff00d0 →			

Onto the Heap!

heap1.cpp			
4	int main() {		
5	int *p = new int;		
6	Sphere *s = new Sphere(10);		
7			
8			
9			
10	return 0;		
11	}		

Stack	Value		Heap	Value
0xffff00f0 →			0x42020 →	
0xffff00e8 →			0x42018 →	
0xffff00e0 →			0x42010 →	
0xffff00d8 →			0x42008 →	
0xffff00d0 →			0x42000 →	

heap2.cpp			
4	int main() {		
5	Sphere *s1 = new Sphere();		
6	Sphere *s2 = s1;		
7			
8	s2->setRadius(10);		
9			
10			
11			
12	return 0;		
13	}		

Stack	Value		Heap	Value
0xffff00f0 →			0x42020 →	
0xffff00e8 →			0x42018 →	
0xffff00e0 →			0x42010 →	
0xffff00d8 →			0x42008 →	
0xffff00d0 →			0x42000 →	

Heap Memory Lifecycle

Memory Address Operators

&s

*ptr

heap-puzzle1.cpp	
1	#include <iostream>
2	using namespace std;
3	
4	int main() {
5	int *p, *q;
6	p = new int;
7	q = p;
8	*q = 8;
9	cout << *p << endl;
10	
11	q = new int;
12	*q = 9;
13	cout << *p << endl;
14	cout << *q << endl;
15	
16	return 0;
17	}

heap-puzzle2.cpp	
1	#include <iostream>
2	using namespace std;
3	
4	int main() {
5	int *x;
6	int size = 3;
7	
8	x = new int[size];
9	
10	for (int i = 0; i < size; i++) {
11	x[i] = i + 3;
12	}
13	
14	delete[] x;
15	}

- | CS 225 – Things To Be Doing: |
|--|
| <ol style="list-style-type: none">1. Work on MP1 – Due Monday, March 2nd at 11:59pm2. Complete first lab – Due Sunday at 11:59pm3. First quiz is next Thursday! See Blackboard for quiz topics. |